

FYS5419/9419,
lecture February
19, 2025

FYS5419/9419 Feb 19

$$H = H_0 + H_1$$

$$H_0 = \begin{bmatrix} E_1 & 0 \\ 0 & E_2 \end{bmatrix}$$

computational basis

$$|0\rangle \quad \text{and} \quad |1\rangle$$
$$\begin{bmatrix} 1 \\ 0 \end{bmatrix} \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

$$H_0 |0\rangle = E_1 |0\rangle \quad \wedge \quad H_0 |1\rangle = E_2 |1\rangle$$

$$H_1 = \begin{bmatrix} V_{11} & V_{12} \\ V_{21} & V_{22} \end{bmatrix}$$

$$H_0 = \frac{E_1 + E_2}{2} \mathbb{1}_{2 \times 2} + \frac{E_1 - E_2}{2} Z$$

$$\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix} \quad \begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$$

$$H_1 = \frac{V_{11} + V_{22}}{2} \mathbb{1} + \frac{V_{11} - V_{22}}{2} Z$$

$$+ V_x X$$

$$V_x = V_{12} = V_{21}$$

$$\begin{bmatrix} 1 & 0 \\ c & 1 \end{bmatrix}$$

$$\langle H \rangle = c \langle \varphi_i | \mathbb{1} | \varphi_i \rangle + d \langle \varphi_i | X | \varphi_i \rangle + e \langle \varphi_i | Z | \varphi_i \rangle$$

Project 1

when we diagonalize

$$|\varphi_1\rangle = \alpha|0\rangle + \beta|1\rangle$$

$$|\varphi_2\rangle = \gamma|0\rangle + \delta|1\rangle$$

initial state $|\varphi\rangle$

$$H|\varphi\rangle = |\varphi'\rangle \cdot \lambda$$

initial state could be $|0\rangle$

and we act on this with
different gates

$|0\rangle \rightarrow ?$

one-qubit system, we can
define a general state on the
Bloch sphere as

$$|\varphi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\varphi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

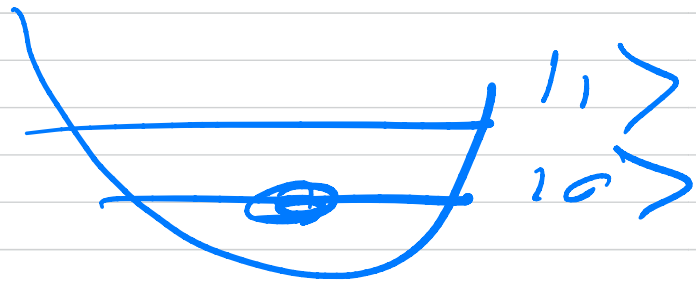
Rotations:

$$R_x(\theta) = e^{i\frac{\theta}{2}X} = \cos\left(\frac{\theta}{2}\right)\mathbb{1}_{2\times 2} - i\sin\left(\frac{\theta}{2}\right)X$$

$$R_y(\theta) = \cos\left(\frac{\theta}{2}\right)\mathbb{1} - i\sin\left(\frac{\theta}{2}\right)Y$$

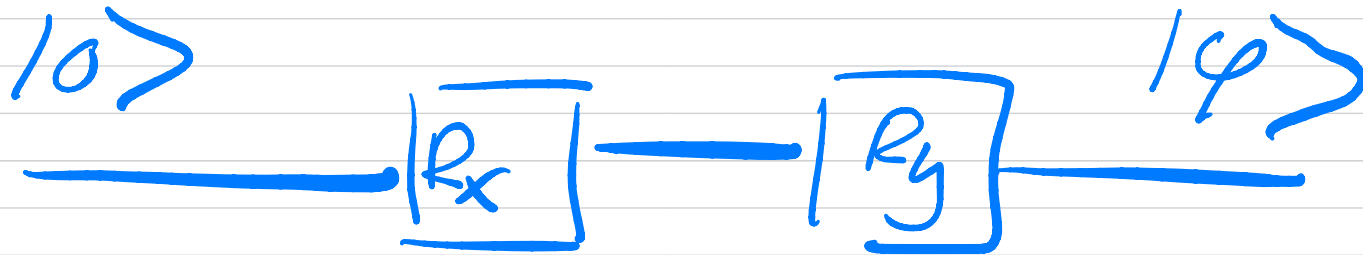
initial state is modified to

$$|\varphi\rangle = R_y(\epsilon) R_x(\phi) |0\rangle = |\varphi(\epsilon, \phi)\rangle$$



$$\langle \varphi | H | \varphi \rangle$$

Graphically !



$$E(\theta, \phi) = \langle \psi(\theta, \phi) | H | \psi(\theta, \phi) \rangle$$

1) optimization part (classical calculation)

$$\frac{\partial E}{\partial \theta} = 0 \quad \wedge \quad \frac{\partial E}{\partial \phi} = 0$$

Gradient descent

2) evaluation of $\langle \psi | H | \psi \rangle$
on a quantum computer

$$\langle \psi | I | \psi \rangle \quad \wedge \quad \langle \psi | Z | \psi \rangle$$

and $\langle \psi | X | \psi \rangle$

1+2 make up the V (Variational)
Q (Quantum) E (Eigensolver)
algorithm

Ideal is to make measurement
in the computational basis

$$\sigma_x |0\rangle = |0\rangle \quad \wedge \quad \sigma_x |1\rangle = |1\rangle$$

$$\sigma_z |0\rangle = |0\rangle \quad \wedge \quad \sigma_z |1\rangle = -|1\rangle$$

$$\sigma_y |0\rangle = |1\rangle \quad \wedge \quad \sigma_y |1\rangle = -|0\rangle$$

Hadamard matrix

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} = \frac{1}{\sqrt{2}} \left(\overset{z}{\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}} + \overset{x}{\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}} \right)$$

$$H|0\rangle = \frac{1}{\sqrt{2}} (|0\rangle + |1\rangle)$$

$$H|1\rangle = \frac{1}{\sqrt{2}} (|0\rangle - |1\rangle)$$

superposition

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} x & z \\ z & x \end{bmatrix} \quad H^+ H = \underline{1}$$

$$H H^+ = \underline{1}$$

$$H^{-1} = H$$

$$Z = HXH \Rightarrow HZH = X$$

$$\frac{1}{\sqrt{2}} [X + Z] \times \frac{1}{\sqrt{2}} [X + Z]$$

$$= \frac{1}{2} \left[\underbrace{X^2}_=1 X + X \underbrace{Z^2}_=1 + Z X \underbrace{Z^2}_=1 + \underbrace{Z X Z}_{-X} \right]$$

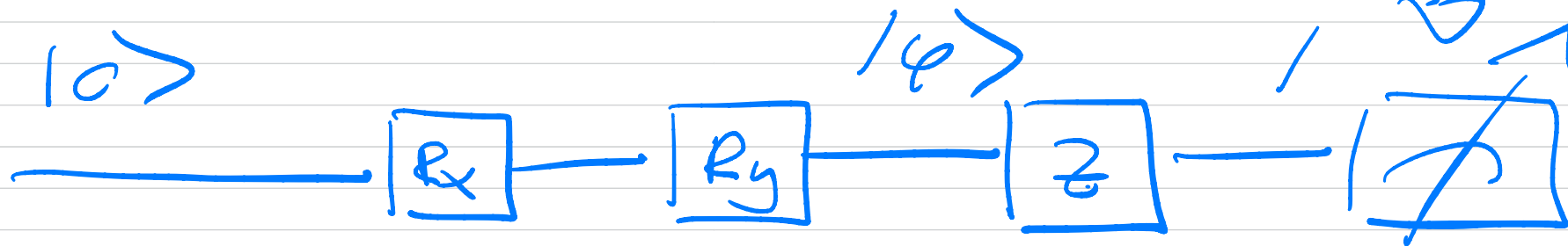
$$= Z$$

$$\langle \varphi | X | \varphi \rangle = \langle \varphi | H Z H | \varphi \rangle$$

$$= \langle \varphi H | Z | H \varphi \rangle = \langle \varphi' | Z | \varphi' \rangle$$

circuit :

$\langle \varphi | z | \varphi \rangle$:

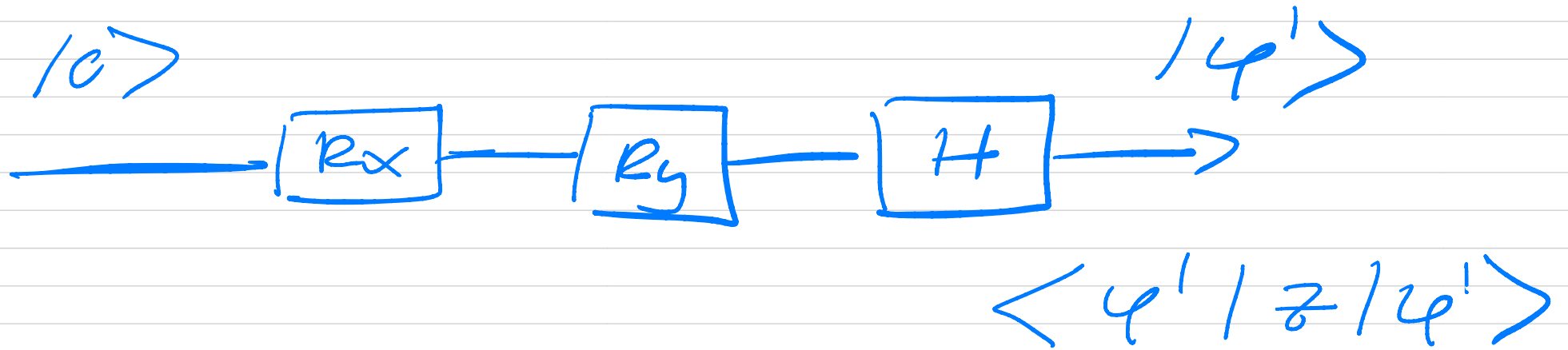
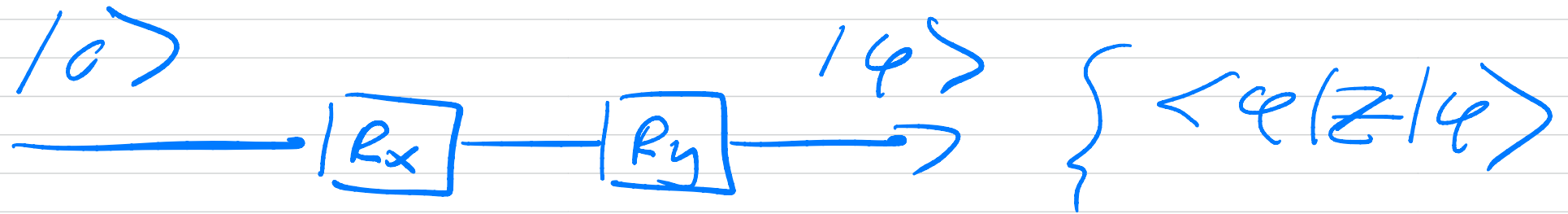


$|0\rangle$



$\langle \varphi | H Z H | \varphi \rangle$

State changes at $z/4$



$$|4\rangle = \alpha |0\rangle + \beta |1\rangle$$

$$P_4 = |4\rangle \langle 4| = |\alpha|^2 |0\rangle \langle 0| +$$

$$\alpha \beta^* |0\rangle\langle 1| + \alpha^* \beta |1\rangle\langle 0|$$

$$+ |\beta|^2 |1\rangle\langle 1|$$

$$= \begin{bmatrix} |\alpha|^2 & \alpha \beta^* \\ \alpha^* \beta & |\beta|^2 \end{bmatrix}$$

$$\text{Tr } \rho = |\alpha|^2 + |\beta|^2 = \underline{1}$$

Entropy

- measure of likelihood.

Example: binary system

$$\text{True} = p \quad p \in [0, 1]$$

$$\text{False} = 1 - p$$

$p = 1$ (True) we know exactly the state of the system.

$$S = -\log_2 p$$

$$p = 1 \quad S = 0$$

requirements

— additive

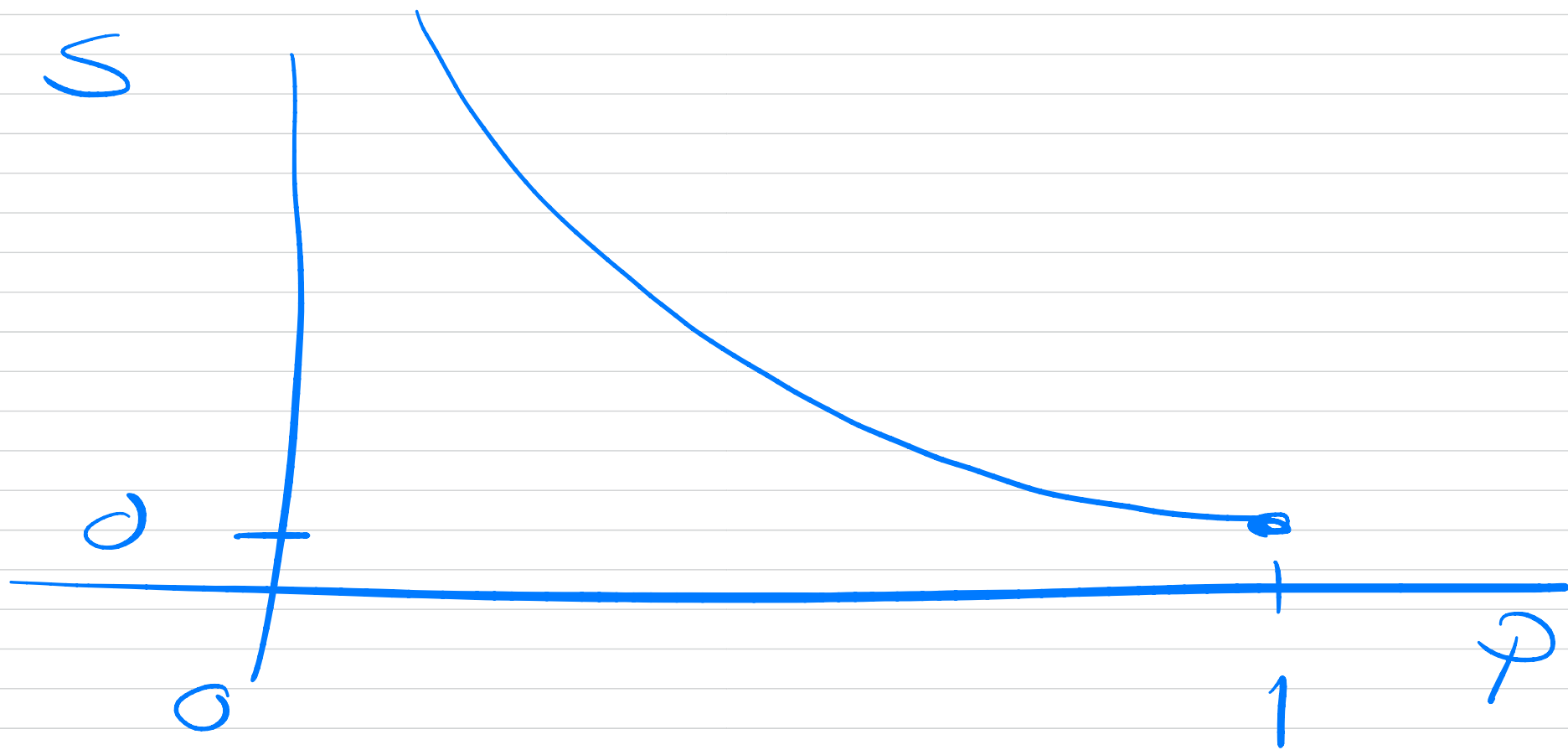
i.i.d.
$$P(x_1, x_2) = P_1(x_1) P_2(x_2)$$
$$= P(x_1) P(x_2)$$

$$S = -\log_2 P(x_1, x_2)$$
$$= -\log_2(P(x_1)) - \log_2(P(x_2))$$

— continuous

— non-negative

$$S = -\log_e p$$



$$p = 0 \Rightarrow \text{False} = \underline{1 - p}$$

$$= \underline{1}$$

Try

$$S = -p \log_2 p \quad \text{in general}$$

$$S(x) = - \sum_{x \in X} P_X(x) \log_2 P_X(x)$$

$$X = \{x_0, x_1, \dots, x_{n-1}\}$$

Binary example

$$S = -(p \log_2 p) - (1-p) \log_2 (1-p)$$
$$p \in [0, 1]$$

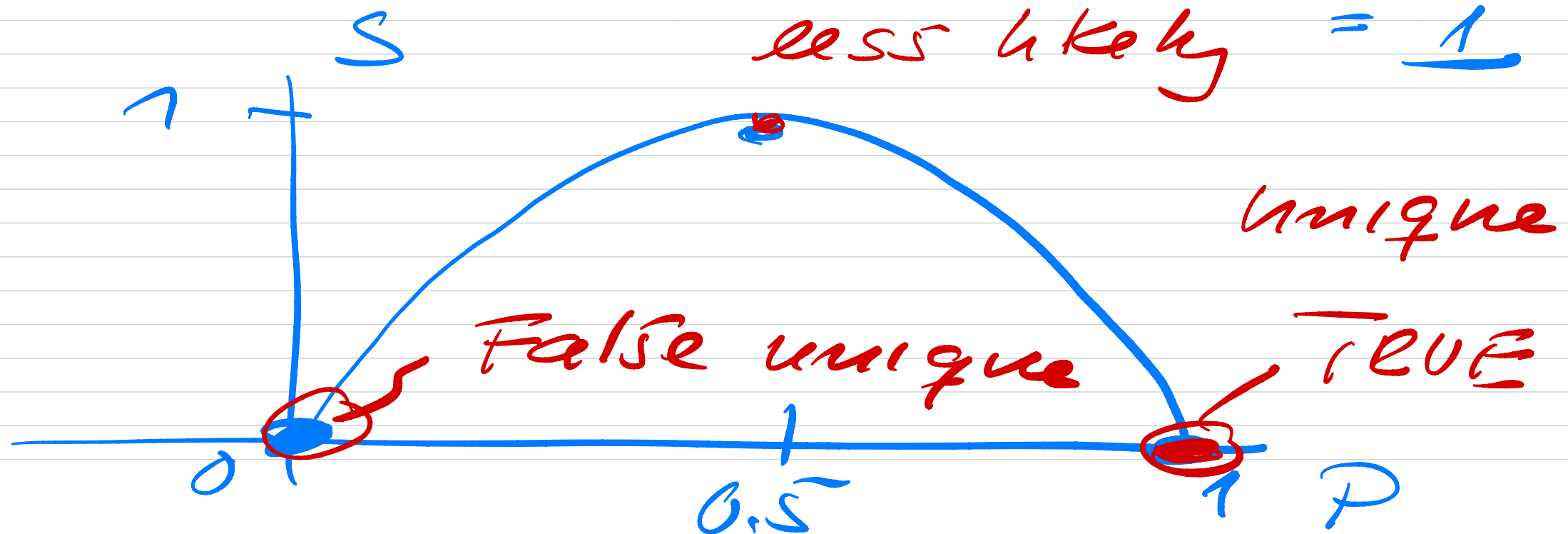
$$p=0 \quad S=0 \quad \text{False (1-p)}$$

$$\left(\lim_{x \rightarrow 0} x \log_2 x = 0 \right)$$

$$p=1 \quad S=0 \quad \text{True}$$

$$p=0.5 \quad S = -\log_2 \frac{1}{2} = \log_2 2$$

less likely = 1



Von Neumann entropy

Density matrix (system A)

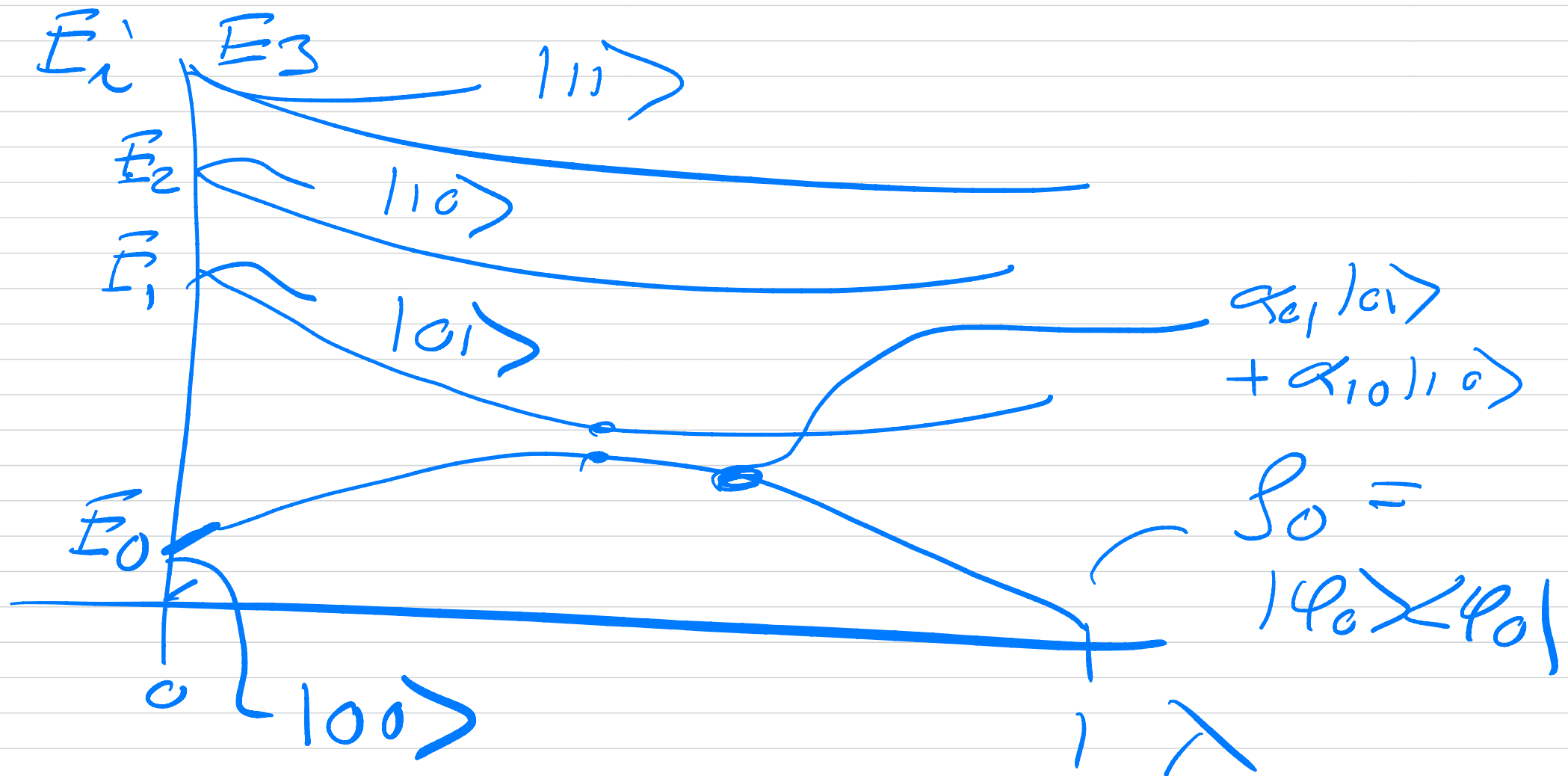
$$\rho_A = \sum_j p_j \underbrace{| \psi_j \rangle_A}_{\text{ONB}} \langle \psi_j |_A$$

Von-Neumann entropy

$$- \text{Tr} [\rho_A \log_2 \rho_A]$$

$$H = H_0 + \lambda H_1$$

$$\lambda = [0, 1]$$



$$|\psi_0\rangle = \alpha_{00}|00\rangle + \alpha_{01}|01\rangle + \alpha_{10}|10\rangle + \alpha_{11}|11\rangle$$